Programming Fundamentals

SEMESTER PROJECT

Made By :

1. **Abdul Wahab (I-CSA-9250017)**
2. **Sajjad Hussain (I-CSA-9250005)**
3. **Adil Hussain (I-CSA-9250018)**

wahabofficial209@gmail.com

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Project Title :

Super Market Billing System + Inventory Management

Project Features :

- Add / Edit / Delete Items
- Display Items
- Search Items
- Create Bills for Customers
- Add Multiple Items in Bill
- Quantity Based Discounts
- Continuous - Non Stoppable Billing
- Category Based Items
- Multiple Menus
- Item Availability Indicator
- Low Stock Remainder

Functions Used :

1. void main_Menu();
2. void billing_Menu();
3. void inventory_Menu();
4. void billing();
5. void billing_History();
6. void add_Items();
7. void remove_Items();
8. void modify_Items();
9. float discounts(int);
10. void view_Items();
11. void low_Stock();
12. void load_Items();
13. void save_All_Items();
14. void search_Items();
15. void save_Billing_History(double total_price);
16. void about();
17. void about_Submenu();
18. void version_About();

void main_Menu();

Code :

```
void main_Menu()
{
   int choice;
   cout<<endl;
         cout<<"Enter 1 to login for Biling."<<endl;
   cout<<"Enter 2 to login as Inventory Manager (Password Required)."<<endl;
   cout<<"Enter 3 to View About."<<endl;
   cout<<"Enter 4 to Exit."<<endl;
   cout<<"Enter Your Choice : ";
   cin>>choice;
   cout<<endl;
   switch(choice)
   {
      case 1:
      {
         billing_Menu();
         break;
      }
      case 2:
      {
         int pass;
         const int pin=1122;
         cout<<"Enter Your Password : ";
         cin>>pass;
         cout<<endl;
         if(pass==pin)
         {
            inventory_Menu();
         }
         else
         {
            cout<<"Wrong Password"<<endl;
            main_Menu();
         }
         break;
      }
      case 3:
      {
         about();
         break;
      }
      case 4:
      {
          cout<<"Successfully Exited."<<endl;
```

```
            break;
                }
        default:
        {
            cout<<"Wrong Input"<<endl;
            main_Menu();
        }
    }
}
```

Output :

```
=======Welcome to Super Market Manager=======


Enter 1 to login for Biling.
Enter 2 to login as Inventory Manager (Password Required).
Enter 3 to View About.
Enter 4 to Exit.
Enter Your Choice :
```

void billing_Menu();

Code:

```
void billing_Menu()
{
    int choice;
    cout<<endl;
    cout<<"========== BILLING MENU ==========="<<endl;
    cout<<"Enter 1 to Start Billing."<<endl;
    cout<<"Enter 2 to View Billing History."<<endl;
    cout<<"Enter 3 to Return to Main Menu."<<endl;
    cout<<"Enter Your Choice : ";
    cin>>choice;
    cout<<endl;
    switch(choice)
    {
        case 1:
        {
            billing();
            break;
        }
```

```
        case 2:
        {
            billing_History();
            break;
        }
        case 3:
        {
            main_Menu();
            break;
        }
        default:
        {
            cout<<"Wrong Input"<<endl;
            main_Menu();
        }
    }
}
```

Output:

```
========== BILLING MENU ==========
Enter 1 to Start Billing.
Enter 2 to View Billing History.
Enter 3 to Return to Main Menu.
Enter Your Choice :
```

void inventory_Menu();

Code :

```
void inventory_Menu()
{
    int choice;
    cout<<endl;
    cout<<"========= INVENTORY MENU =========="<<endl;
    cout<<"Enter 1 to Add Items."<<endl;
    cout<<"Enter 2 to Remove Items."<<endl;
    cout<<"Enter 3 to Modify Items."<<endl;
    cout<<"Enter 4 to View Items."<<endl;
    cout<<"Enter 5 to View Low Stock."<<endl;
    cout<<"Enter 6 to Search Between Items."<<endl;
    cout<<"Enter 7 to Return to Main Menu."<<endl;
```

```
    cout<<"Enter Your Choice : ";
    cin>>choice;
    cout<<endl;
    switch(choice)
    {
        case 1:
                {
                        add_Items();
                        break;
                }
        case 2:
                {
                        remove_Items();
                        break;
                }
        case 3:
                {
                        modify_Items();
                        break;
                }
        case 4:
                {
                        view_Items();
                        break;
                }
        case 5:
                {
                        low_Stock();
                        break;
                }
        case 6:
                {
                        search_Items();
                        break;
                }
        case 7:
                {
                        main_Menu();
                        break;
                }
        default:
                {
                        main_Menu();
                        break;
                }
    }
}
```

Output :

```
Enter Your Password : 1122


========= INVENTORY MENU =========
Enter 1 to Add Items.
Enter 2 to Remove Items.
Enter 3 to Modify Items.
Enter 4 to View Items.
Enter 5 to View Low Stock.
Enter 6 to Search Between Items.
Enter 7 to Return to Main Menu.
Enter Your Choice : _
```

void billing();

Code :

```
void billing()
{
    int stop=0;
    int id;
    int qty;
    double total_price=0;
    cout<<"======= ENTERING BILLING MODE =======">>endl;
    cout<<"Type -1 to finish billing"<<endl<<endl;
    while(stop!=-1)
    {
        cout<<"Enter Item ID : ";
        cin>>stop;
        cout<<endl;
        if(stop==-1)
        {
            break;
        }
                id=stop;
        bool found=false;
        for(int i=0;i<item_count;i++)
        {
            if(item[i].id==id)
            {
```

```
                found=true;
                cout<<left<<setw(20)<<"Name: "<<item[i].name<<endl;
                cout<<left<<setw(20)<<"Category: "<<item[i].category<<endl;
                cout<<left<<setw(20)<<"Price: "<<fixed<<setprecision(2)<<item[i].price<<endl;
                cout<<left<<setw(20)<<"Available Quantity: "<<item[i].quantity<<endl<<endl;
                cout<<"Enter Quantity to Buy: ";
                cin>>qty;
                cout<<endl;
                if(qty<=0)
                {
                   cout<<"Invalid Quantity"<<endl;
                }
                else if(qty>item[i].quantity)
                {
                   cout<<"Not Enough Stock"<<endl;
                }
                else if(qty>=5)
                {
                        float discount=discounts(qty);
                        total_price=(total_price+(item[i].price*qty))*discount;
                        cout<<"Discount of Excess Quantity Applied."<<endl;
                                        }
                else
                {
                   total_price=total_price+(item[i].price*qty);
                   item[i].quantity=item[i].quantity-qty;
                   cout<<"Item Added to Bill"<<endl;
                }
                cout<<endl;
                break;
             }
          }
          if(found==false)
          {
             cout<<"Item Not Found"<<endl<<endl;
          }
       }
       fstream fout("items.txt", ios::out);
       for(int i=0;i<item_count;i++)
       {
          fout<<item[i].id<<" "<<item[i].name<<" "<<item[i].category<<" "<<item[i].price<<"
"<<item[i].quantity<<endl;
       }
       fout.close();
       fstream bill("billing_history.txt", ios::app);
       bill<<"-----------------------------"<<endl;
       bill<<"New Bill Generated"<<endl;
```

```
    bill<<"Total Bill Amount: "<<fixed<<setprecision(2)<<total_price<<endl;
    bill<<"============================"<<endl;
    bill.close();
    cout<<"======= FINAL BILL ======="<<endl;
    cout<<"Total Amount: "<<fixed<<setprecision(2)<<total_price<<endl;
    cout<<endl;
        bool another_bill;
        cout<<"For Another Bill Press 1 and 0 to Exit Billing"<<endl;
        cout<<"Enter Your Choice : ";
        cin>>another_bill;
        cout<<endl;
        if(another_bill==1 || another_bill==0)
        {
            if(another_bill==1)
            {
                billing();
            }
            else
            {
                cout<<"Exiting Billing....."<<endl;
                main_Menu();
            }
        }
}
```

Output :

```
======= ENTERING BILLING MODE =======
Type -1 to finish billing

Enter Item ID : 1

Name:                 Milk
Category:             Drink
Price:                200.00
Available Quantity:   86

Enter Quantity to Buy: 5

Discount of Excess Quantity Applied.

Enter Item ID : 2

Name:                 Rice
Category:             Food
Price:                300.00
Available Quantity:   98

Enter Quantity to Buy: 1

Item Added to Bill

Enter Item ID : -1

======= FINAL BILL =======
Total Amount: 800.00

For Another Bill Press 1 and 0 to Exit Billing
Enter Your Choice : 0

Exiting Billing.....
```

void billing_History();

Code :

```
void billing_History()
{
    fstream fin("billing_history.txt", ios::in);
    if(!fin)
    {
        cout<<"No Billing History Found"<<endl;
        billing_Menu();
        return;
    }
    string line;
    cout<<"\n======= BILLING HISTORY =======\n"<<endl;
    while(getline(fin,line))
    {
        cout<<line<<endl;
    }
```

```
    fin.close();
    cout<<endl;
    billing_Menu();
}
```

Code :

```
======= BILLING HISTORY =======

-----------------------------
New Bill Generated
Total Bill Amount: 0
=============================
-----------------------------
New Bill Generated
Total Bill Amount: 1000
=============================
-----------------------------
New Bill Generated
Total Bill Amount: 400
=============================
-----------------------------
New Bill Generated
Total Bill Amount: 0
=============================
-----------------------------
New Bill Generated
Total Bill Amount: 2000.00
=============================
-----------------------------
New Bill Generated
Total Bill Amount: 720.00
=============================
-----------------------------
New Bill Generated
Total Bill Amount: 800.00
=============================
```

void add_Items();

Code :

```
void add_Items()
{
    int n;
    cout<<"Enter total number of items to add: ";
    cin>>n;
    cout<<endl;
    for(int i=0;i<n;i++)
    {
        cout<<"Enter Item ID: ";
        cin>>item[item_count].id;
        cout<<"Enter Item Name: ";
        cin>>item[item_count].name;
        cout<<"Enter Item Category: ";
        cin>>item[item_count].category;
        cout<<"Enter Item Price: ";
        cin>>item[item_count].price;
        cout<<"Enter Item Quantity: ";
        cin>>item[item_count].quantity;
```

```
        cout<<endl;
        item_count++;
    }
    save_All_Items();
    inventory_Menu();
}
```

Output :

```
Enter total number of items to add: 1

Enter Item ID: 3
Enter Item Name: Chocolate
Enter Item Category: Food
Enter Item Price: 50
Enter Item Quantity: 100
```

void remove_Items();

Code :

```
void remove_Items()
{
    int id;
    cout<<"Enter Item ID to Remove: ";
    cin>>id;
    cout<<endl;
    for(int i=0;i<item_count;i++)
    {
        if(item[i].id==id)
        {
            for(int j=i;j<item_count-1;j++)
            {
                item[j]=item[j+1];
            }
            item_count--;
            break;
        }
    }
    cout<<"Item Removed Successfully!!!"<<endl;
    save_All_Items();
    inventory_Menu();
}
```

Output :

```
Enter Item ID to Remove: 4

Item Removed Successfully!!!
```

void modify_Items();

Code :

```
void modify_Items()
{
    int id;
    cout<<"Enter Item ID to Modify: ";
    cin>>id;
    cout<<endl;
    for(int i=0;i<item_count;i++)
    {
        if(item[i].id==id)
        {
            cout<<"Enter New Price: ";
            cin>>item[i].price;
            cout<<"Enter New Quantity: ";
            cin>>item[i].quantity;
            cout<<endl;
            break;
        }
    }
    cout<<"Item Modified Successfully.!!!"<<endl;
    save_All_Items();
    inventory_Menu();
}
```

Output :

```
Enter Item ID to Modify: 1

Enter New Price: 210
Enter New Quantity: 90

Item Modified Successfully.!!!
```

float discounts(int);

Code :

```
float discounts(int q)
{
        float disc;
        disc=q*0.10;
        return disc;
}
```

void view_Items();

Code :

```
void view_Items()
{

cout<<setw(5)<<"ID"<<setw(20)<<"Name"<<setw(20)<<"Category"<<setw(10)<<"Price"<<set
w(10)<<"Qty"<<endl;
   cout<<setw(65)<<"-----------------------------------------------------------"<<endl;
   for(int i=0;i<item_count;i++)
   {
      cout<<setw(5)<<item[i].id<<setw(20)<<item[i].name<<setw(20)<<item[i].category
         <<setw(10)<<fixed<<setprecision(2)<<item[i].price<<setw(10)<<item[i].quantity<<endl;
   }
   cout<<endl;
   inventory_Menu();
}
```

Output :

```
Enter Your Choice : 4

   ID                 Name            Category     Price       Qty
    -----------------------------------------------------------
    1                 Milk               Drink    210.00        90
    2                 Rice                Food    300.00        97
    3            Chocolate                Food     50.00       100
```

void low_Stock();

Code :

```
void low_Stock()
{
   cout<<endl;
   cout<<setw(20)<<"Low Stock Items"<<endl;
   cout<<setw(30)<<"--------------------------"<<endl;
```

```
    for(int i=0;i<item_count;i++)
    {
        if(item[i].quantity<5)
        {
            cout<<setw(20)<<item[i].name<<setw(10)<<item[i].quantity<<endl;
        }
    }
    cout<<endl;
    inventory_Menu();
}
```

Output :

```
Low Stock Items
--------------------------
```

(Currently Nothing Stocked Low)

void load_Items();

Code :

```
void load_Items()
{
    fstream fin("items.txt", ios::in);
            if(!fin)
    {
        return;
    }
            while(fin>>item[item_count].id>>item[item_count].name>>item[item_count].category>
>item[item_count].price>>item[item_count].quantity)
    {
        item_count++;
        if(item_count>=all_items)
        {
            break;
        }
    }
    fin.close();
}
```

Output :

Items getting loaded in every function.

void save_All_Items();

Code :

```
void save_All_Items()
{
    fstream fout("items.txt", ios::out);
    for(int i=0;i<item_count;i++)
    {
        fout<<item[i].id<<" "<<item[i].name<<" "<<item[i].category<<" "<<item[i].price<<"
"<<item[i].quantity<<endl;
    }
    fout.close();
}
```

Output :

Items being saved after changings.

void search_Items();

Code :

```
void search_Items()
{
        int searchId;

        cout<<endl;
        cout<<"========= SEARCH ITEM =========="<<endl;
        cout<<"Enter Item ID to Search : ";
        cin>>searchId;
        cout<<endl;

        for(int i=0;i<item_count;i++)
        {
                if(item[i].id==searchId)
                {
                        cout<<"Item Found Successfully"<<endl;
                        cout<<"---------------------------------------------"<<endl;

                        cout<<left<<setw(20)<<"Item ID:"<<item[i].id<<endl;
                        cout<<left<<setw(20)<<"Name:"<<item[i].name<<endl;
                        cout<<left<<setw(20)<<"Category:"<<item[i].category<<endl;

        cout<<left<<setw(20)<<"Price:"<<fixed<<setprecision(2)<<item[i].price<<endl;
                        cout<<left<<setw(20)<<"Quantity:"<<item[i].quantity<<endl;

                        cout<<"---------------------------------------------"<<endl;
                        cout<<endl;

                        inventory_Menu();
```

```
                return;
            }
        }

        cout<<"Item Not Found!"<<endl<<endl;
        inventory_Menu();
    }
```

Output :

```
========= SEARCH ITEM =========
Enter Item ID to Search : 1

Item Found Successfully
---------------------------------------------
Item ID:            1
Name:               Milk
Category:           Drink
Price:              210.00
Quantity:           90
---------------------------------------------
```

void save_Billing_History(double total_price);

Code :

```
void save_Billing_History(double total_price)
{
   fstream fout("billing_history.txt", ios::app);
   fout<<"-----------------------------"<<endl;
   fout<<"New Bill Generated"<<endl;
   fout<<"Total Bill Amount: "<<total_price<<endl;
   fout<<"=============================="<<endl;
   fout.close();
}
```

Output :

Saving Billing History After Billing.(File Handling)

void about();

Code :

```
void about()
{
```

```
        int choice;
        cout<<endl;
        cout<<"Press 1 for Program's Info."<<endl;
        cout<<"Press 2 for Developer's Info"<<endl;
        cout<<"Press 3 to Go Back."<<endl;
        cout<<endl;
        cout<<"Enter Your Choice : ";
        cin>>choice;
        cout<<endl;
        switch(choice)
        {
                case 1:
                        {
                                version_About();
                                break;
                        }
                case 2:
                        {
                                about_Submenu();
                                break;
                        }
                case 3:
                        {
                                main_Menu();
                                break;
                        }
                default:
                        {
                                cout<<endl;
                                cout<<"Wrong Input \n";
                                cout<<endl;
                                about();
                                break;
                        }
        }
}
```

Output :

```
Press 1 for Program's Info.
Press 2 for Developer's Info
Press 3 to Go Back.

Enter Your Choice :
```

void about_Submenu();

Code :

```
void about_Submenu()
{
	int choice;
	cout<<"Enter 1 to Show Developers Name."<<endl;
	cout<<"Enter 2 to Show their Roll Numbers."<<endl;
	cout<<"Enter 3 to Show their System Id's."<<endl;
	cout<<"Enter 4 to Contact Us."<<endl;
	cout<<"Enter 5 to Return Back."<<endl;
	cout<<endl;
	cout<<"Enter Your Choice : ";
	cin>>choice;
	cout<<endl;
	switch(choice)
	{
		case 1:
			{
				cout<<"This Program is Developed By : "<<endl;
				cout<<"1.  Abdul Wahab"<<endl;
				cout<<"2.  Adil Hussain"<<endl;
				cout<<"3.  Sajjad Hussain"<<endl;
				cout<<endl;
				about_Submenu();
				break;
			}
		case 2:
			{
				cout<<"Roll Numbers are :"<<endl;
				cout<<"1. Abdul Wahab : I-CSA-9250017"<<endl;
				cout<<"2. Adil Hussain : I-CSA-9250018"<<endl;
				cout<<"3. Sajjad Hussain : I-CSA-9250005"<<endl;
				cout<<endl;
				about_Submenu();
				break;
			}
		case 3:
			{
				cout<<"System Id's are : "<<endl;
				cout<<"1. Abdul Wahab : NUML-ISB-F25-15632"<<endl;
				cout<<"2. Adil Hussain : NUML-ISB-F25-15635"<<endl;
				cout<<"3. Sajjad Hussain : NUML-ISB-F25-07771"<<endl;
				cout<<endl;
				about_Submenu();
```

```
                break;
            }
        case 4:
            {
                cout<<"Contact Us Through Emails :"<<endl;
                cout<<"1. wahabofficial209@gmail.com"<<endl;
                cout<<"2. adilnhussain512@gmail.com"<<endl;
                cout<<"3. sajjadhussainnn65@gmail.com"<<endl;
                cout<<endl;
                about_Submenu();
                break;
            }
        case 5:
            {
                about();
                break;
            }
        default:
            {
                cout<<"Wrong Input\n";
                cout<<endl;
                about_Submenu();
                break;
            }
    }
}
```

Output :

```
Enter Your Choice : 1

This Program is Developed By :
1.  Abdul Wahab
2.  Adil Hussain
3.  Sajjad Hussain

Enter 1 to Show Developers Name.
Enter 2 to Show their Roll Numbers.
Enter 3 to Show their System Id's.
Enter 4 to Contact Us.
Enter 5 to Return Back.

Enter Your Choice : 2

Roll Numbers are :
1. Abdul Wahab : I-CSA-9250017
2. Adil Hussain : I-CSA-9250018
3. Sajjad Hussain : I-CSA-9250005

Enter 1 to Show Developers Name.
Enter 2 to Show their Roll Numbers.
Enter 3 to Show their System Id's.
Enter 4 to Contact Us.
Enter 5 to Return Back.

Enter Your Choice : 3

System Id's are :
1. Abdul Wahab : NUML-ISB-F25-15632
2. Adil Hussain : NUML-ISB-F25-15635
3. Sajjad Hussain : NUML-ISB-F25-07771

Enter 1 to Show Developers Name.
Enter 2 to Show their Roll Numbers.
Enter 3 to Show their System Id's.
Enter 4 to Contact Us.
Enter 5 to Return Back.

Enter Your Choice : 4

Contact Us Through Emails :
1. wahabofficial209@gmail.com
2. adilnhussain512@gmail.com
3. sajjadhussainnn65@gmail.com
```

void version_About();

Code :

```
void version_About()
{
        cout<<"Version Number : 1"<<endl;
        cout<<"Functions Contain : 18"<<endl;
        cout<<"Lines Count : ~600"<<endl;
        cout<<endl;
        about();
```

```
}
```

Output :

```
Version Number : 1
Functions Contain : 19
Lines Count : ~600
```

Project Code :

```
#include<iostream>
#include<string>
#include<fstream>
#include<iomanip>

using namespace std;

void main_Menu();
void billing_Menu();
void inventory_Menu();
void billing();
void billing_History();
void add_Items();
void remove_Items();
void modify_Items();
float discounts(int);
void view_Items();
void low_Stock();
void load_Items();
void save_All_Items();
void search_Items();
void save_Billing_History(double total_price);
void about();
void about_Submenu();
void version_About();

struct items
{
    double price;
    int id;
    string name;
    string category;
    int quantity;
```

```
};

const int all_items=50;
items item[all_items];
int item_count=0;

int main()
{
   load_Items();
   cout<<"\n=======Welcome to Super Market Manager=======\n"<<endl;
   main_Menu();
   return 0;
}

void main_Menu()
{
   int choice;
   cout<<endl;
          cout<<"Enter 1 to login for Biling."<<endl;
   cout<<"Enter 2 to login as Inventory Manager (Password Required)."<<endl;
   cout<<"Enter 3 to View About."<<endl;
   cout<<"Enter 4 to Exit."<<endl;
   cout<<"Enter Your Choice : ";
   cin>>choice;
   cout<<endl;
   switch(choice)
   {
      case 1:
      {
         billing_Menu();
         break;
      }
      case 2:
      {
         int pass;
         const int pin=1122;
         cout<<"Enter Your Password : ";
         cin>>pass;
         cout<<endl;
         if(pass==pin)
         {
            inventory_Menu();
         }
         else
         {
            cout<<"Wrong Password"<<endl;
            main_Menu();
```

```
            }
            break;
        }
        case 3:
        {
            about();
            break;
        }
        case 4:
        {
             cout<<"Successfully Exited."<<endl;
            break;
                        }
        default:
        {
            cout<<"Wrong Input"<<endl;
            main_Menu();
        }
    }
}

void billing_Menu()
{
    int choice;
    cout<<endl;
    cout<<"========== BILLING MENU ==========="<<endl;
    cout<<"Enter 1 to Start Billing."<<endl;
    cout<<"Enter 2 to View Billing History."<<endl;
    cout<<"Enter 3 to Return to Main Menu."<<endl;
    cout<<"Enter Your Choice : ";
    cin>>choice;
    cout<<endl;
    switch(choice)
    {
        case 1:
        {
            billing();
            break;
        }
        case 2:
        {
            billing_History();
            break;
        }
        case 3:
        {
            main_Menu();
```

```
            break;
        }
        default:
        {
            cout<<"Wrong Input"<<endl;
            main_Menu();
        }
    }
}

void inventory_Menu()
{
    int choice;
    cout<<endl;
    cout<<"========= INVENTORY MENU =========="<<endl;
    cout<<"Enter 1 to Add Items."<<endl;
    cout<<"Enter 2 to Remove Items."<<endl;
    cout<<"Enter 3 to Modify Items."<<endl;
    cout<<"Enter 4 to View Items."<<endl;
    cout<<"Enter 5 to View Low Stock."<<endl;
    cout<<"Enter 6 to Search Between Items."<<endl;
    cout<<"Enter 7 to Return to Main Menu."<<endl;
    cout<<"Enter Your Choice : ";
    cin>>choice;
    cout<<endl;
    switch(choice)
    {
        case 1:
                        {
                                add_Items();
                                break;
                        }
        case 2:
            {
                                remove_Items();
                                break;
                        }
        case 3:
                        {
                                modify_Items();
                                break;
                        }
        case 4:
                        {
                                view_Items();
                                break;
                        }
```

```
        case 5:
                        {
                                low_Stock();
                                break;
                        }
        case 6:
                        {
                                search_Items();
                                break;
                        }
                case 7:
                        {
                                inventory_Menu();
                                break;
                        }
        default:
                        {
                                main_Menu();
                                break;
                        }
    }
}

void billing()
{
    int stop=0;
    int id;
    int qty;
    double total_price=0;
    cout<<"======= ENTERING BILLING MODE =======">>endl;
    cout<<"Type -1 to finish billing"<<endl<<endl;
    while(stop!=-1)
    {
        cout<<"Enter Item ID : ";
        cin>>stop;
        cout<<endl;
        if(stop==-1)
        {
            break;
        }
                id=stop;
        bool found=false;
        for(int i=0;i<item_count;i++)
        {
            if(item[i].id==id)
            {
                found=true;
```

```
                cout<<left<<setw(20)<<"Name: "<<item[i].name<<endl;
                cout<<left<<setw(20)<<"Category: "<<item[i].category<<endl;
                cout<<left<<setw(20)<<"Price: "<<fixed<<setprecision(2)<<item[i].price<<endl;
                cout<<left<<setw(20)<<"Available Quantity: "<<item[i].quantity<<endl<<endl;
                cout<<"Enter Quantity to Buy: ";
                cin>>qty;
                cout<<endl;
                if(qty<=0)
                {
                   cout<<"Invalid Quantity"<<endl;
                }
                else if(qty>item[i].quantity)
                {
                   cout<<"Not Enough Stock"<<endl;
                }
                else if(qty>=5)
                {
                        float discount=discounts(qty);
                        total_price=(total_price+(item[i].price*qty))*discount;
                        cout<<"Discount of Excess Quantity Applied."<<endl;
                                        }
                else
                {
                   total_price=total_price+(item[i].price*qty);
                   item[i].quantity=item[i].quantity-qty;
                   cout<<"Item Added to Bill"<<endl;
                }
                cout<<endl;
                break;
             }
          }
          if(found==false)
          {
             cout<<"Item Not Found"<<endl<<endl;
          }
       }
       fstream fout("items.txt", ios::out);
       for(int i=0;i<item_count;i++)
       {
          fout<<item[i].id<<" "<<item[i].name<<" "<<item[i].category<<" "<<item[i].price<<"
"<<item[i].quantity<<endl;
       }
       fout.close();
       fstream bill("billing_history.txt", ios::app);
       bill<<"-----------------------------"<<endl;
       bill<<"New Bill Generated"<<endl;
       bill<<"Total Bill Amount: "<<fixed<<setprecision(2)<<total_price<<endl;
```

```
    bill<<"============================="<<endl;
    bill.close();
    cout<<"======= FINAL BILL ======="<<endl;
    cout<<"Total Amount: "<<fixed<<setprecision(2)<<total_price<<endl;
    cout<<endl;
            bool another_bill;
            cout<<"For Another Bill Press 1 and 0 to Exit Billing"<<endl;
            cout<<"Enter Your Choice : ";
            cin>>another_bill;
            cout<<endl;
            if(another_bill==1 || another_bill==0)
            {
                    if(another_bill==1)
                    {
                            billing();
                    }
                    else
                    {
                            cout<<"Exiting Billing....."<<endl;
                            main_Menu();
                    }
            }
}

void load_Items()
{
    fstream fin("items.txt", ios::in);
            if(!fin)
    {
        return;
    }
            while(fin>>item[item_count].id>>item[item_count].name>>item[item_count].category>
>item[item_count].price>>item[item_count].quantity)
    {
        item_count++;
        if(item_count>=all_items)
        {
            break;
        }
    }
    fin.close();
}

void save_All_Items()
{
    fstream fout("items.txt", ios::out);
    for(int i=0;i<item_count;i++)
```

```
    {
        fout<<item[i].id<<" "<<item[i].name<<" "<<item[i].category<<" "<<item[i].price<<"
"<<item[i].quantity<<endl;
    }
    fout.close();
}

void add_Items()
{
    int n;
    cout<<"Enter total number of items to add: ";
    cin>>n;
    cout<<endl;
    for(int i=0;i<n;i++)
    {
        cout<<"Enter Item ID: ";
        cin>>item[item_count].id;
        cout<<"Enter Item Name: ";
        cin>>item[item_count].name;
        cout<<"Enter Item Category: ";
        cin>>item[item_count].category;
        cout<<"Enter Item Price: ";
        cin>>item[item_count].price;
        cout<<"Enter Item Quantity: ";
        cin>>item[item_count].quantity;
        cout<<endl;
        item_count++;
    }
    save_All_Items();
    inventory_Menu();
}

void remove_Items()
{
    int id;
    cout<<"Enter Item ID to Remove: ";
    cin>>id;
    cout<<endl;
    for(int i=0;i<item_count;i++)
    {
        if(item[i].id==id)
        {
            for(int j=i;j<item_count-1;j++)
            {
                item[j]=item[j+1];
            }
            item_count--;
```

```
            break;
        }
         cout<<"Item Removed Successfully!!!"<<endl;
    }
    save_All_Items();
    inventory_Menu();
}

void modify_Items()
{
    int id;
    cout<<"Enter Item ID to Modify: ";
    cin>>id;
    cout<<endl;
    for(int i=0;i<item_count;i++)
    {
        if(item[i].id==id)
        {
            cout<<"Enter New Price: ";
            cin>>item[i].price;
            cout<<"Enter New Quantity: ";
            cin>>item[i].quantity;
            cout<<endl;
            break;
        }
    }
    cout<<"Item Modified Successfully.!!!"<<endl;
    save_All_Items();
    inventory_Menu();
}

void view_Items()
{

cout<<setw(5)<<"ID"<<setw(20)<<"Name"<<setw(20)<<"Category"<<setw(10)<<"Price"<<set
w(10)<<"Qty"<<endl;
    cout<<setw(65)<<"-----------------------------------------------------------------"<<endl;
    for(int i=0;i<item_count;i++)
    {
        cout<<setw(5)<<item[i].id<<setw(20)<<item[i].name<<setw(20)<<item[i].category
            <<setw(10)<<fixed<<setprecision(2)<<item[i].price<<setw(10)<<item[i].quantity<<endl;
    }
    cout<<endl;
    inventory_Menu();
}

void low_Stock()
```

```
{
    cout<<endl;
    cout<<setw(20)<<"Low Stock Items"<<endl;
    cout<<setw(30)<<"--------------------------"<<endl;
    for(int i=0;i<item_count;i++)
    {
        if(item[i].quantity<5)
        {
            cout<<setw(20)<<item[i].name<<setw(10)<<item[i].quantity<<endl;
        }
    }
    cout<<endl;
    inventory_Menu();
}

void billing_History()
{
    fstream fin("billing_history.txt", ios::in);
    if(!fin)
    {
        cout<<"No Billing History Found"<<endl;
        billing_Menu();
        return;
    }
    string line;
    cout<<"\n======= BILLING HISTORY =======\n"<<endl;
    while(getline(fin,line))
    {
        cout<<line<<endl;
    }
    fin.close();
    cout<<endl;
    billing_Menu();
}

void save_Billing_History(double total_price)
{
    fstream fout("billing_history.txt", ios::app);
    fout<<"-----------------------------"<<endl;
    fout<<"New Bill Generated"<<endl;
    fout<<"Total Bill Amount: "<<total_price<<endl;
    fout<<"============================="<<endl;
    fout.close();
}

float discounts(int q)
{
```

```
        float disc;
        disc=q*0.10;
        return disc;
}

void about()
{
        int choice;
        cout<<endl;
        cout<<"Press 1 for Program's Info."<<endl;
        cout<<"Press 2 for Developer's Info"<<endl;
        cout<<"Press 3 to Go Back."<<endl;
        cout<<endl;
        cout<<"Enter Your Choice : ";
        cin>>choice;
        cout<<endl;
        switch(choice)
        {
                case 1:
                        {
                                version_About();
                                break;
                        }
                case 2:
                        {
                                about_Submenu();
                                break;
                        }
                case 3:
                        {
                                main_Menu();
                                break;
                        }
                default:
                        {
                                cout<<endl;
                                cout<<"Wrong Input \n";
                                cout<<endl;
                                about();
                                break;
                        }
        }
}

void about_Submenu()
{
        int choice;
```

```
cout<<"Enter 1 to Show Developers Name."<<endl;
cout<<"Enter 2 to Show their Roll Numbers."<<endl;
cout<<"Enter 3 to Show their System Id's."<<endl;
cout<<"Enter 4 to Contact Us."<<endl;
cout<<"Enter 5 to Return Back."<<endl;
cout<<endl;
cout<<"Enter Your Choice : ";
cin>>choice;
cout<<endl;
switch(choice)
{
	case 1:
		{
			cout<<"This Program is Developed By : "<<endl;
			cout<<"1.  Abdul Wahab"<<endl;
			cout<<"2.  Adil Hussain"<<endl;
			cout<<"3.  Sajjad Hussain"<<endl;
			cout<<endl;
			about_Submenu();
			break;
		}
	case 2:
		{
			cout<<"Roll Numbers are :"<<endl;
			cout<<"1. Abdul Wahab : I-CSA-9250017"<<endl;
			cout<<"2. Adil Hussain : I-CSA-9250018"<<endl;
			cout<<"3. Sajjad Hussain : I-CSA-9250005"<<endl;
			cout<<endl;
			about_Submenu();
			break;
		}
	case 3:
		{
			cout<<"System Id's are : "<<endl;
			cout<<"1. Abdul Wahab : NUML-ISB-F25-15632"<<endl;
			cout<<"2. Adil Hussain : NUML-ISB-F25-15635"<<endl;
			cout<<"3. Sajjad Hussain : NUML-ISB-F25-07771"<<endl;
			cout<<endl;
			about_Submenu();
			break;
		}
	case 4:
		{
			cout<<"Contact Us Through Emails :"<<endl;
			cout<<"1. wahabofficial209@gmail.com"<<endl;
			cout<<"2. adilnhussain512@gmail.com"<<endl;
			cout<<"3. sajjadhussainnn65@gmail.com"<<endl;
```

```
                        cout<<endl;
                        about_Submenu();
                        break;
                }
            case 5:
                {
                        about();
                        break;
                }
            default:
                {
                        cout<<"Wrong Input\n";
                        cout<<endl;
                        about_Submenu();
                        break;
                }
        }
}

void version_About()
{
        cout<<"Version Number : 1"<<endl;
        cout<<"Functions Contain : 18"<<endl;
        cout<<"Lines Count : ~600"<<endl;
        cout<<endl;
        about();
}

void search_Items()
{
        int searchId;

        cout<<endl;
        cout<<"========= SEARCH ITEM ========="<<endl;
        cout<<"Enter Item ID to Search : ";
        cin>>searchId;
        cout<<endl;

        for(int i=0;i<item_count;i++)
        {
                if(item[i].id==searchId)
                {
                        cout<<"Item Found Successfully"<<endl;
                        cout<<"---------------------------------------------"<<endl;

                        cout<<left<<setw(20)<<"Item ID:"<<item[i].id<<endl;
                        cout<<left<<setw(20)<<"Name:"<<item[i].name<<endl;
```

```
                    cout<<left<<setw(20)<<"Category:"<<item[i].category<<endl;

        cout<<left<<setw(20)<<"Price:"<<fixed<<setprecision(2)<<item[i].price<<endl;
                    cout<<left<<setw(20)<<"Quantity:"<<item[i].quantity<<endl;

                    cout<<"---------------------------------------------"<<endl;
                    cout<<endl;

                    inventory_Menu();
                    return;
                }
        }

        cout<<"Item Not Found!"<<endl<<endl;
        inventory_Menu();
}
```